

Junk Science and Deceptive Publications: Implications for the Biomedical Community

The world of academic publishing has evolved substantially in the past decades, as the Open Access movement has pushed to make data and research more freely available. Today there are two ways in which publishing costs are covered. The first is the more traditional “reader pays model” in which publishers earn their revenue through subscription fees that are charged to libraries and individual users. The second model, highly influenced by the Open Access movement, is the “author pays model”, where the researcher pays for their research to be published, either out of their own pocket or via grant funding (Seethapathy et al., 2016). There are many advantages to the “author pays model”, including making research more freely available to the public, a move which could be beneficial to many researchers, particularly those from developing countries who may not have as much monetary resources to spend on high-cost research publications; however, the dark side of the Open Access movement includes the operations of “predatory publishing”, journals that exist for profit and publish research that does not meet standard publishing standards (Sharma & Verma, 2018). This darker side of the journalism industry has some potential to be quite damaging, particularly in the biomedical industry. This paper will examine predatory publishing and the open access movement in relation to the biomedical research community. A discussion of predatory publishing will be prefaced by a discussion of the introduction of the Open Access movement and its influence on traditional publishing practices. After, there will be a discussion of which sorts of researchers are engaging in predatory publishing and their motivations to do so. Finally, this paper will look at the implications of predatory publishing on the biomedical community as well as potential ways in which this sort of negative publishing can be handled in the future.

The Open Access (OA) movement has been highly influential in the world of publishing. The main idea of OA is that researchers (or the grants that fund their research) should bear the brunt of publishing fees associated with publishing research in academic journals. This takes away the financial burden from the reader to the author (Yeo-Teh & Tang, n.d.). Major advantages include increased access to knowledge by researchers and students worldwide and increased visibility and readership (Sharma). For the world of biomedical science, increased access to knowledge could be highly beneficial to boosting potentially life-saving research (Sharma & Verma, 2018). For authors who are publishing in an OA journal, there are many associated fees including an article-processing fee and other fees for the editing and peer review process (Georg et al., 2023). There are, however, many advantages of publishing research in a reputable OA journal. Authentic journals have functional editorial offices and editorial boards that facilitate appropriate peer review (Yeo-Teh & Tang, n.d.). Authentic journals also offer higher visibility in comparison to traditional journals, and they tend to have a shorter publication process as well as high likelihoods of citations (Livas & Delli, 2023). For some researchers, publishing in OA journals is risky as they feel that OA journals do not have the same

standards as traditional academic journals (Sonne et al., 2020). Regardless, the Open Access movement has transformed the publishing world.

The term “predatory journals” was first used in an article published to PubMed in 2012 via a note in Nature journal by Prof. Jeffrey Beall (Forero et al., 2018). Prof. Beall gained notoriety for maintaining an online list of potential and possible predatory publishers and journals. Some of the criteria that Prof. Beall noted for this list included immense amounts of email spam asking for article submissions, lack and/or misrepresentation of the publisher’s headquarters’ location, and an absence of copyediting in published articles. The last version of this list was published in 2017, with 1,155 publishers and 1,294 standalone journals being included (Forero et al., 2018). A conference that met in Ottawa defined the phenomenon of “predatory journals” as,

“Entities that prioritize self-interest at the expense of scholarship and are characterized by false or misleading information, deviation from best editorial and publication practices, a lack of transparency, and/or the use of aggressive and indiscriminative solicitation practices” (Grudniewicz et al., 2019).

As of 2023, the number of predatory publications was estimated to exceed 15,000 titles; however, it can be hard to count the exact number as many journals operate in a “grey zone between fraud and legitimacy” (Livas & Delli, 2023). Following the OA model, predatory journals advertise their publications with promises of high acceptance rates and fast-tracked publishing. Publishing research often in a few days or weeks, many of these journals do not offer the traditional peer-review process, meaning that the research they publish is often not vetted. Some journals lie to their customers about impact possibilities and publish research that may not receive any citations. These journals are also characterized by their absence of an editorial office or association with a well-recognized institution and/or organization (Forero et al., 2018). Many predatory journals are operated by one person and one PC! The result of predatory journals is a myriad of publications that would never be published in a traditionally peer-reviewed journal. Even if the research being published in a predatory journal was legitimately well-done, appropriate research, often these journals are encrypted and inaccessible to search engines (Sharma & Verma, 2018). Some predatory journals only exist for short amounts of time, and when they disappear, all of their content and published reports disappear (Georg et al., 2023). All of these factors accumulate into potential losses of scientific data and funding. For genuine researchers, this means a loss of countless hours of hard work.

Many who write about the phenomenon of predatory publishing ask whether those who publish in these sorts of journals are aware that they are engaging with predatory journals. It is noted that scholars from developing countries are at the highest risk of becoming victims to these practices, as OA journals are not affordable for the majority of individuals (Forero et al., 2018; Livas & Delli, 2023; Ng & Haynes, 2021). Others may publish in these journals because of the “publish or perish” mentality,

which refers to a desire either to see one's name in writing or to advance their careers, as many academics are required to have a certain number of publications to move forward with their doctoral studies or gain tenure (Forero et al., 2018; Yeo-Teh & Tang, n.d.). The top five countries that are targeted and publish in predatory journals are India, the U.S.A., Iran, Indonesia, and Turkey (Forero et al., 2018). Seethapathy et.al's study focuses on those who have engaged in predatory publishing in India, sending out a questionnaire to gauge how aware individuals were that they were engaging in the practice. India ranks 13th in the world for high-quality publications and their research outputs have grown substantially since 2012. However, India is still a major contributor of articles in either poor quality or predatory journals. In response to Seethapathy et.al.'s questionnaire, 80% of the people who responded noted that they were unaware that the journals they had published in were predatory. Further, 90% of the authors noted that they had paid for the article processing charges from their own pockets and not via research grants or institutional funds. Further, many of the respondents considered research publications as a major achievement and felt academic pressure to publish as larger numbers of publications gives job securities and promotions (Seethapathy et al., 2016). Seethapathy et.al writes, "Most authors are not necessarily tricked into publishing in predatory journals; they probably submit to them are aware of the circumstances and take a calculated risk that experts who evaluate their publication lists will not bother to check the journal credentials in detail. Hence, we do not uncritically see the authors are unknowing victims" (Seethapathy et al., 2016). Noting that researchers who publish in predatory journals are often "knowing victims" can help outline which steps need to be taken to decrease the amount of research that is published in these sorts of journals. It is understood that many young researchers feel the pressure of meeting requirements that state a solid number of publications for advancement, potentially academic boards can develop alternative benchmarks for advancement from high numbers of publications.

The biomedical community is at acute risk in relation to inadequate research being published in predatory journals. This sort of publishing has direct implications on healthcare and research, which can have global impacts. As there is no way to verify the quality of articles and avoid harmful findings, some of this research can be directly harmful to patients and others in the field. Further, this research directly undermines the credibility of previous scientific literature that has been published in the health sciences (Forero et al., 2018; Ng & Haynes, 2021). Uncredible publications are being included in systematic reviews that directly inform the development of clinical practice guidelines and often clinicians and policy makers depend on scientific research for decision-making (Ng & Haynes, 2021). The result of this is that individuals can no longer blindly trust a scientific biomedical journal (Sharma & Verma, 2018). It is not only clinicians and policy makers who are reading these sorts of publications, as patients can search the internet for information about their illness and have the potential to learn misinformation from this sort of "junk science" (Forero et al., 2018). While there are many, potentially devastating, impacts of predatory publishing to the biomedical community, there are ways in which the

community has influenced higher numbers of these sorts of publications. Specifically, many Medical Councils have requirements that researchers have a certain number of publications before they are considered for promotion (Forero et al., 2018). For those that feel the pressure from needing a higher number of publications, predatory publishing is an easy way to advance in the biomedical field. This directly influences the concept of “citation contamination” which is the propagation of flawed and unvetted science. This diminishes the importance and impact of genuine and legitimate publications (Yeo-Teh & Tang, n.d.). Further, some members of the biomedical community may be unaware that they are interacting with predatory journals, potentially unaware of their existence in the first place (Ng & Haynes, 2021). Placing an emphasis on educating the biomedical community about the predatory publishing industry and its direct and very negative impacts on the hard work that biomedical professionals do could be an important start.

There are a variety of ways in which researchers and academics can gain awareness of the sorts of predatory journals that exist in their field as well as ways in which larger institutions can lead actions against these sorts of journals. There have been multiple lists that attempt to showcase which journals are predatory. Most famous of these lists is Beall’s list, which was an online list of possible, potential, or likely predatory publishers and journals. Originally started in 2012, the list was maintained by Prof. Beall until 2017 where it was taken down due to pressure from publishers, journal editors, and many universities. Even though the list was taken down, many institutions still use the list to vet out their researchers’ publications (Sharma & Verma, 2018). Prof. Beall never supported a regular body to oversee scholarly publishers but instead promoted education and scientific literacy of all researchers to recognize these sorts of journals, noting that this would be balanced with an ethical duty to publish research that has been appropriately vetted by a peer review process. While many universities are still using the list, many are worried that the disappearance of Beall’s list has increased in the number of articles published in predatory journals (Forero et al., 2018). There have been few studies on the accuracy of Beall’s list for noting and identifying predatory publishers, with Georg et.al’s study being among the few. Georg et.al find that 82.6% ISSN journals on Beall’s list were not found in any investigated and quality driven databases, showing that they were highly likely to be predatory. This study showed that many of the journals on Beall’s list were receiving citations, which shows that publications that are likely published in predatory journals without a lengthy peer-review process have been used for further research (Georg et al., 2023). Predatory journals can publish articles on all sorts of subjects, meaning that all areas of research could be contaminated with worthless research unknowingly.

Another list of predatory publications that has been developed is Cabell’s Blacklist which utilizes a 65-point list in order to identify predatory publishers. It is noted, however, that many of the points in the 65-point list are misleading or inaccurate. This had led to the blacklisting of low-quality

journals that may not have malicious intent (Ng & Haynes, 2021). Many of the checklists that are published to identify predatory journals lack concrete guidelines on how to be used and often use binary cut-offs with the items in the checklists in order to identify whether a journal is predatory or not. Ng and Haynes (2021) notes that many of the content in these checklists do not differ much from each other, and that there is major work that needs to be done to compile a composite checklist that eliminates non-discriminated items.

While Beall and Cabell's list have been instrumental in bringing awareness to the predatory publishing industry, without robust checklists and indices these sorts of lists have the potential of being pushed to the side as "unreliable" if it remains impossible to fully check if the journals they are noting as "predatory" are legitimate or not. There has been a proposed predatory publishing rank metric called the "Predatory Rate" (PR). This would be based on 14 criteria; however, it is noted that more robust indices would be instrumental in screening predatory publishing (Forero et al., 2018). Another list that has been developed is an international initiative called "Think. Check. Submit." (Forero et al., 2018; Sharma & Verma, 2018). This initiative gives multiple recommendations and questions to researchers and potential authors so they can evaluate their intentions before publishing. The "Think. Check. Submit." initiative has a central website published in dozens of languages that provides a checklist that researchers can fill out with downloadable results. Some of the questions provided in the checklist include, "Is the journal clear about the type of peer review it uses?", "Can you easily identify and contact the publisher?", "Are articles indexed and/or archived in dedicated services?", and "Is the publisher a current member of a recognized industry initiative?" (*Think. Check. Submit.*, n.d.). Research universities and departments can make a large difference in the amount of research being published in predatory journals by requiring that their staff complete and provide a passing score from the "Think. Check. Submit." checklist before submitting their research to any academic journal.

Some authors argue that calling some publications "predatory journals" does not paint a full picture of the entire phenomenon in the publishing industry. Eriksson and Helgesson (2018) argue that this side of the publishing industry should be divided into two realms, each with their own set of criteria: deceptive journals and low-quality journals. Deceptive journals are the publications that deceive authors, readers and institutions by misrepresenting and publishing misleading information about the peer review and editing process prospective research would be subject to. This includes journals that hide where their editorial board is situated and lie about facts such as costs to publish and indexing services provided (Eriksson & Helgesson, 2018). Low-quality journals, on the other hand, publish research of poor quality or may have too broad of a scope of interest. These publications may have no strategy regarding misconduct or digital preservation and their websites may lack important information regarding the peer review process and location of headquarters (Eriksson & Helgesson, 2018). While low-quality journals are harmful in their own way, they are not actively lying and

attempting to deceive researchers. They also tend not to spam researchers with adverts in an attempt to receive research and they do not target researchers from developing countries. The main way to coerce low-quality journals into taking publishing more seriously is to rank them low in relation to other more strict journals on lists that consider a holistic approach. Eriksson and Helgesson (2018) argues, however, that deceptive journals need to be met with legal action for “violating research integrity”. The main difference between the proposed labels of “low quality-journal” and “deceptive journal” is intention. Being a journal that is not rigorous or having low criteria for publishing does not necessarily entail that the publication is “deceptive” or “predatory”, the journal is mostly undiscerning.

One last potential for scanning and identifying “predatory publications” is with the use of AI-driven tools such as ChatGPT, a natural language processing application that generates text that reads almost human-like in response to the prompts the user provides. There has been research to generate website classification through deep learning attempts, however, this is in its infancy and must be done with caution as training the algorithm incorrectly could lead to mis-classification of legitimate journals. Al-Moghrabi et.al explains that this process of searching through websites and journals to find ones that may be “predatory” is called “website scraping”. This process uses indicators such as grammatical and spelling mistakes, existence of different languages on the same page, claims in the paper concerning metrics or indexing, and/or the absence of copyright policy or publishing license. Potential risks of using ChatGPT for “website scraping” to find predatory publishers include running into biased content due to the training sets the algorithm uses, legal concerns regarding data privacy, and the potential future generation of false scientific content which may resemble as sound data to non-experts (Al-Moghrabi et al., 2024). Al-Moghrabi et.al’s research showed that there is a high potential for ChatGPT to identify predatory journals; however, they are hesitant to give too much control the natural language processing software as it is unclear if predatory journals are also putting in their own safeguards to deceive artificial intelligence. The processor may have differing levels of performance when applied to journals in differing fields (Al-Moghrabi et al., 2024). Further research into the potential of using artificial intelligence to differentiate between Erikssons and Helgesson’s proposed split between “deceptive journal” and “low-quality journals” could be highly influential for creating lists that aid researchers in their goals to publishing research in high quality journals.

Predatory publishing is an industry that has a far reach and negative influence within the wider publishing world. Its effects are generally unquantifiable in the biomedical world as metrics to study the effects of research that has not gone through a rigorous peer review process on both future research and decisions that medical communities make in creating guidelines and patient care is an extremely large scope. Regardless, there is a consensus that predatory publishing has been harmful to the biomedical research community. Main ways to combat this negative trend is through the creation and use of checklists to vet out predatory journals as well as the potential development of more advanced

use of natural language processors to vet out journals that are deceptive in nature. Multiple authors noted that there has been a development tied to the predatory publishing world, which is the development of “fake” or “predatory” conferences (Sharma & Verma, 2018; Sonne et al., 2020). These are conferences that are organized by these predatory publishers in an attempt to be scholarly. They are,

“Low quality academic meetings [that have] the primary aim of making money and not supporting [scholastic development]; there is no effective peer review, allowing anyone to purchase a speaking slot; the organizer employs deceit, the most common forms being peer review, hiding the company headquarters’ true location, and concealing the for-profit nature of the company” (McCrostie, 2018).

South Korea became the first country to ban its researchers from attending these sorts of predatory conferences, requiring their academics to provide lengthy information regarding the conferences they are attending that must be approved of by the government (Sonne et al., 2020). Potentially universities can start developing screening policies for these predatory conferences and create lists similar to Beall’s list.

Bibliography

- Al-Moghrabi, D., Abu Arqub, S., Maroulakos, M. P., Pandis, N., & Fleming, P. S. (2024). Can ChatGPT identify predatory biomedical and dental journals? A cross-sectional content analysis. *Journal of Dentistry*, 142, 104840.
- Eriksson, S., & Helgesson, G. (2018). Time to stop talking about “predatory journals.” *Learned Publishing*, 31(2), 181–183.
- Forero, D. A., Oermann, M. H., Manca, A., Deriu, F., Mendieta-Zerón, H., Dadkhah, M., Bhad, R., Deshpande, S. N., Wang, W., & Cifuentes, M. P. (2018). Negative Effects of “Predatory” Journals on Global Health Research. *Annals of Global Health*, 84(4), 584–589.
<https://doi.org/10.29024/aogh.2389>
- Georg, R., Berger, M., Koeller, M., Richtig, M., Richtig, E., Sheffel, J., Maurer, M., & Siebnhaar, F. (2023). Predatory journals: Perception, impact and use of Beall’s list by the scientific community—A bibliometric big data study. *PLoS One*, 18(7), 1–17.
<https://doi.org/10.1371/journal.pone.0287547>
- Grudniewicz, A., Moher, D., Cobey, K. D., Bryson, G. L., Cukier, S., Allen, K., Ardern, C., Blacom, L., Barros, T., Berger, M., Ciro, J. B., Cugusi, L., Donaldson, M. R., Egger, M., Graham, I. D., Hodgkinson, M., Khan, K. M., Mabizela, M., Manca, A., ... Lalu, M. M. (2019). Predatory journals: No definition, no defence. *Nature*, 572(210), 2.
- Livas, C., & Delli, K. (2023). “Dear Doctor, greetings of the day!”: A 1-year observational study of presumed predatory journal invitations. *Progress in Orthodontics*, 24(1), 21.
<https://doi.org/10.1186/s40510-023-00471-6>
- McCrostie, J. (2018). Predatory Conferences: A case of academic cannibalism. *International Higher Education*, 93(8), 6–8.

- Ng, J. Y., & Haynes, B. R. (2021). “Evidence-based checklists” for identifying predatory journals have not been assessed for reliability or validity: An analysis and proposal for moving forward. *Journal of Clinical Epidemiology*, 138, 40–48. <https://doi.org/10.1016/j.jclinepi.2021.06.015>
- Seethapathy, G. S., Santhosh Kumar, J. U., & Hareesha, A. S. (2016). India’s scientific publication in predatory journals: Need for regulating quality of Indian science and education. *Current Science*, 111(11), 1759–1765. <https://doi.org/10.18520/cs/v111/i11/1759-1764>
- Sharma, H., & Verma, S. (2018). Predatory journals: The rise of worthless biomedical science. *Journal of Postgraduate Medicine*, 64(4).
- Sonne, C., Ok, Y. S., Lam, S. S., Rinklebe, J., Alstrup, A. K. O., & Kim, K.-H. (2020). First predatory journals, now conferences: The need to establish lists of fake conferences. *The Science of the Total Environment*, 715, 136990. <https://doi.org/10.1016/j.scitotenv.2020.136990>
- Think. Check. Submit. (n.d.). [Website.]. <https://thinkchecksubmit.org/journals/>
- Yeo-Teh, N. S. L., & Tang, B. L. (n.d.). Wilfully submitting to and publishing in predatory journals—A covert form of research misconduct? *Biochem Med*, 31(3), 1–8. <https://doi.org/10.11613/BM.2021.030201>